

Hypothesis Testing and the Module 2 Deliverable

Module 2, Part 3 (October 16)

QBA 5330

A Different Question

- Part 2 asked: *is the true national rate at least leadership's minimum bar?*
- New question: *has the new storefront's performance actually changed from the old storefront's history?*
- That's a **hypothesis test** question.

Testing the Conversion Rate

$H_0 : p = 0.12$ (historical rate) vs. $H_a : p \neq 0.12$

```
1 p_hat <- 0.15; p0 <- 0.12; n <- 200
2
3 z_stat <- (p_hat - p0) / sqrt(p0 * (1 - p0) / n)
4 p_value <- 2 * (1 - pnorm(abs(z_stat)))
5 c(z = z_stat, p_value = p_value)
```

```
z    p_value
1.3055824 0.1916946
```

At $\alpha = 0.05$: fail to reject H_0 .

Testing the Order Value

$H_0 : \mu = 65$ (historical average) vs. $H_a : \mu \neq 65$

```
1 x_bar <- 71.50; mu0 <- 65; s <- 42; n <- 25
2
3 t_stat <- (x_bar - mu0) / (s / sqrt(n))
4 p_value <- 2 * pt(-abs(t_stat), df = n - 1)
5 c(t = t_stat, p_value = p_value)
```

```
      t      p_value
0.7738095 0.4465993
```

At $\alpha = 0.05$: fail to reject H_0 .

Three Lenses on One Pilot

Question	Tool	This pilot
Best single guess?	Point estimate	$\hat{p} = 0.15; \bar{x} = \71.50
Plausible range?	Confidence interval	$(0.101, 0.199);$ (\$54.16, \$88.84)
Clears leadership's bar?	CI vs. threshold	Conversion: barely. Order value: not clearly.
Changed from history?	Hypothesis test	Fail to reject, both

The Honest Takeaway

No clean “yes, launch” or “no, don’t.”

Good analytics **narrows the range of reasonable disagreement**. It doesn’t always resolve it.

The Module 2 Deliverable

- A short **risk and uncertainty** exercise, using the same three-lens approach:
 1. Expected value / probability
 2. Confidence interval
 3. Hypothesis test
- Applied to a scenario involving **demand, quality, conversion, or customer behavior**, calculated in Excel and reproduced or extended in R.

Looking Ahead to Saturday

- Today: quantifying uncertainty.
- Tomorrow (10/17): **comparing groups and evaluating causal claims**, using the Google Project Oxygen case.